

Teacher Pack: Short Diagnostic Pre-Test

Source reference: Singapore MOE *Primary Mathematics Syllabus* (2021 syllabus for P1-P6, updated October 2025). This short version samples the main P1-P4 strands in 20 questions.

B. Teacher Answer Key

Question	Answer	Skill Category	Approx. Level	What It Checks
1	11	Number Sense and Place Value	P1	Continues a simple number sequence.
2	74	Number Sense and Place Value	P1	Compares two-digit numbers.
3	80	Number Sense and Place Value	P2	Understands place value in a 3-digit number.
4	13	Addition and Subtraction	P1	Adds within 20.
5	7	Addition and Subtraction	P1	Subtracts within 20.
6	73	Addition and Subtraction	P2	Adds two 2-digit numbers.
7	325	Addition and Subtraction	P2/P3	Subtracts 3-digit numbers with regrouping.
8	12	Multiplication and Division Readiness	P1	Understands multiplication as equal groups.
9	42	Multiplication and Division Readiness	P3	Recalls multiplication facts beyond the 5-times table.
10	7 each, 1 left	Multiplication and Division Readiness	P3	Solves division with remainder.
11	1	Fractions	P3	Finds an equivalent fraction.
12	$\frac{3}{4}$	Fractions	P3	Compares unlike fractions.
13	\$2.30	Money and Time	P2	Writes money in decimal notation.
14	\$1.80	Money and Time	P3	Finds change using money subtraction.
15	45 minutes	Money and Time	P3	Finds duration between two

16	200	Measurement	P3	times. Converts metres to centimetres.
17	6 cm	Measurement	P4	Finds a missing rectangle dimension from area.
18	4	Geometry and Shapes	P4	Knows a rectangle's angle property.
19	4	Data and Graphs	P2/P3	Reads a picture graph with scale and compares two values.
20	15	Word Problems and Mathematical Reasoning	P4	Finds a fraction of a set.

C. Scoring Sheet

1 point per question. Total: 20 points.

Skill Category	Number of Questions	Question Numbers	Student Score	Notes
Number Sense and Place Value	3	1-3	___ / 3	Check counting sequence, comparison, place value.
Addition and Subtraction	4	4-7	___ / 4	Check facts and written computation.
Multiplication and Division Readiness	3	8-10	___ / 3	Check equal groups, multiplication and division facts.
Fractions	2	11-12	___ / 2	Check part-whole understanding and comparison.
Money and Time	3	13-15	___ / 3	Check money notation, change, and clock time.
Measurement	2	16-17	___ / 2	Check unit conversion and perimeter.
Geometry and Shapes	1	18	___ / 1	Check angle and shape properties.
Data and Graphs	1	19	___ / 1	Check graph reading and comparison.
Word Problems and	1	20	___ / 1	Check applying fraction ideas in

Mathematical
Reasoning

context.

Total **20** **1-20** **___ / 20**

Use total score
together with
category profile.

D. Overall Score Interpretation

Overall Score	Placement Meaning	Teaching Interpretation
0-5 points	Beginning / Emerging	Likely needs early number sense, counting, shapes, and simple addition support.
6-10 points	Lower Primary Developing	Likely working around P1-P2 level. Needs support with place value, addition/subtraction, money, time, and early multiplication.
11-15 points	Middle Primary Developing	Likely working around P2-P3 level. Can handle some basic operations but may need support with fractions, measurement, and word problems.
16-18 points	Upper Primary Ready	Likely working around P3-P4 level. Ready for more complex operations and applications.
19-20 points	Advanced / Needs Enrichment	Likely ready for stronger P4-style extension and challenge work.

E. Skill Category Interpretation

Number Sense and Place Value

- 0-40%: Student needs support with counting, comparing numbers, and place value.
- 41-70%: Student has partial place value understanding but still makes errors.
- 71-100%: Student is secure and ready for larger numbers and more advanced operations.

Addition and Subtraction

- 0-40%: Student needs support with basic facts and simple addition/subtraction.
- 41-70%: Student can solve some sums but may make regrouping or place value errors.
- 71-100%: Student is secure with core addition and subtraction skills.

Multiplication and Division Readiness

- 0-40%: Student needs concrete work on equal groups, sharing, and repeated addition.
- 41-70%: Student understands the ideas but may be weak in fact recall.
- 71-100%: Student is ready for stronger multiplication and division work.

Fractions

- 0-40%: Student needs support with fractions as equal parts of a whole.
- 41-70%: Student understands some fraction ideas but may confuse comparison.
- 71-100%: Student is ready for broader fraction work.

Money and Time

- 0-40%: Student needs support with coins, money notation, and reading time.
- 41-70%: Student can solve familiar tasks but may struggle with change or elapsed time.
- 71-100%: Student is secure with practical money and time tasks.

Measurement

- 0-40%: Student needs support with units and measuring ideas.
- 41-70%: Student shows some understanding but may make conversion or perimeter errors.
- 71-100%: Student is ready for more advanced measurement work.

Geometry and Shapes

- 0-40%: Student needs support naming and recognising shapes.
- 41-70%: Student recognises some shapes but may be unsure about properties.
- 71-100%: Student is secure with basic geometry recognition.

Data and Graphs

- 0-40%: Student needs support reading simple picture graphs.
- 41-70%: Student can read some graph information but may miss scale.
- 71-100%: Student reads simple data displays accurately.

Word Problems and Mathematical Reasoning

- 0-40%: Student needs support choosing the correct operations in context.
- 41-70%: Student can solve some routine problems but may lose track in multi-step tasks.
- 71-100%: Student applies operations appropriately in context.

F. Recommended Grouping Plan

Group	Suggested Profile	Teaching Focus
Group 1: Foundations	Students with very low overall scores or weak number and operation basics	Counting, number order, number bonds, simple addition/subtraction, shape recognition.
Group 2: Lower Primary Skills	Students with partial basic skills but clear gaps in place value, money, time, and early multiplication	Place value, addition/subtraction fluency, money, time, graph reading, multiplication readiness.
Group 3: Middle Primary Skills	Students mostly secure with basics and ready for fractions, measurement, and word problems	Multiplication/division, fractions, perimeter, money, time, applied problem solving.
Group 4: Extension Group	Students with very high scores and mostly secure	Harder word problems, stronger fraction work,

understanding across
strands

area/perimeter, and richer
reasoning tasks.

G. Short Teacher Notes on How to Use the Pre-Test

1. This shorter version works well as a 20-25 minute screen or quick placement check.
2. Mark by category as well as by total score.
3. If a student does poorly overall, follow up with oral counting, number recognition, and concrete materials.
4. This revised short version has a stronger middle and upper range than the first draft, but the 40-question version still gives finer placement detail.